

Solution Requirements

Date	27 June 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine.
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviours (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Soil & Environmental Data Input	The system allows users to enter Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall values required for crop prediction.	High
2	Input Validation	The system validates all entered values to ensure complete and valid data before processing the prediction.	High

3	Machine Learning Prediction	The system analyzes the input data using the trained machine learning model and predicts the most suitable crop.	High
4	Crop Suitability Analysis	The system evaluates whether the selected soil and environmental conditions are suitable for the recommended crop.	High
5	Recommendation Dashboard	The application displays the recommended crop along with prediction results in a clear and user-friendly interface.	High
6	Business Rules	The recommendation is generated only after all mandatory input parameters are provided and fall within valid ranges supported by the trained model.	High
7	Agricultural Insights	The system provides crop-related insights to help users understand the recommendation and support better farming decisions.	Medium
8	Future Enhancements	The application is designed to support future features such as fertilizer recommendation, real-time weather integration, and crop disease prediction.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The application should process user input and display the crop recommendation without noticeable delay.	Prediction displayed within 2 seconds
2	Scalability	The system should support additional crop datasets, new features, and increased usage without major changes to the application.	Supports future feature and dataset expansion
3	Security & Data Privacy	User inputs should be validated, and the system should process data securely without storing personal information unnecessarily.	100% input validation with secure data handling
4	Reliability & Availability	The application should consistently generate accurate crop recommendations and remain stable during normal operation.	99% successful prediction rate during testing
5	Usability & Accessibility	The interface should be simple, intuitive, and easy to use for farmers with basic computer knowledge.	Users can complete prediction in 3 simple steps

6	Maintainability & Portability	The project should have a modular code structure and run on Windows, Linux, and macOS with minimal configuration.	Cross-platform support with modular Python code
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